

Galois Theory of the Horizon Cover

Abel–Ruffini obstruction, Kummer lifts, and
wreath-product monodromy at four charges

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Abstract

The inner and outer horizon entropies of a Kerr–Newman black hole are the two roots of a monic quadratic whose coefficients are *rational* in the asymptotic charges: $z^2 - e_1 z + e_2 = 0$ with $e_1 = 2\pi(2M^2 - Q^2)$ and $e_2 = \pi^2(4J^2 + Q^4)$. We develop the thermodynamics of the horizon pair as the Galois theory of this cover. The $\mathbb{Z}/2$ grading by the deck transformation (the horizon swap) reproduces the known left/right thermodynamic dictionary — sector entropies and temperatures, sector first laws and Smarr formulas, the universal area product, the hidden-conformal temperatures and central charge, and extremal chirality — from the single principle that *invariants descend to the observables and odd data does not*. The Galois-odd datum of the pair is the covariant $\Theta = ST$; the temperature itself is mixed under the horizon swap, $T_+ \mapsto T_-$, and the two temperatures form their own conjugate pair with rational symmetric functions. For the four-charge (Cvetič–Youm) generalization the invariant persists, $e_2 = \pi^2(4J^2 + N_1 N_2 N_3 N_4)$ with $N_i = 4Q_i$, but the cover degree over the physical field $\mathbb{Q}(M, Q_1, \dots, Q_4)$ jumps to five. We certify, by Dedekind reduction at multiple rational points, that the Galois group of the seed-mass fiber is S_5 ; consequently no closed-form radical expression exists for the mass fiber, and — via an exact collapse of the radical tower — none exists for the horizon entropies themselves: $(S_+ + S_-)^2/\pi^2$ is a primitive element of the S_5 field, while $S_+ S_-$ remains rational. The solvability threshold sits exactly at four charges. The Galois norm of the mass fiber factors completely into the sixteen BPS-type linear forms $M \pm Q_1 \pm Q_2 \pm Q_3 \pm Q_4$, exhibiting the marginal-stability walls as the vanishing locus of a norm. To our knowledge the algebraic-obstruction layer of these results has not appeared in the literature; a full prior-art search is in progress. We further determine the first decorated covers above the quintic. The static ordered-horizon field has degree 20 and normal-closure group $C_2^2 \text{ wr } S_5$. Rotation preserves that degree and closure group, whereas adjoining an independent axial square root raises the sheet degree to 40 and the normal closure to $C_2^3 \text{ wr } S_5$. A conjugate Kummer-module theorem reduces these conclusions to full-rank parity matrices of divisorial valuations. Finally, we classify tame inertia by colored cycle data and certify the realized branch-incidence scheme in Macaulay2; the rotating difference divisor is prime, reduced, and irreducible on the incidence cover.

1 Introduction

Two families of exact results dominate the thermodynamics of multi-horizon black holes. The first is the left/right (“inner mechanics”) apparatus: sector entropies $S_{L,R} = (S_+ \pm S_-)/2$, sector temperatures, first laws and Smarr formulas on both horizons, and negative inner-horizon temperatures, developed by Cvetič and Larsen [1, 2] and given a systematic Gibbsian formulation by

Cvetič, Gibbons, Lü and Pope [3]. The second is the universal area product: S_+S_- is independent of the mass and plausibly quantized [4, 5, 6], a fact used as evidence for a conformal-field-theory description of the microstates in the hidden-conformal program [7, 8, 9].

Both families are computed, throughout the literature, by *manipulating the roots*: one writes S_+ and S_- explicitly in an auxiliary parametrization (the solution-generating frame: a seed mass m and boosts δ_i [11, 12]) and forms the combinations of interest. This paper starts from an elementary observation about that method:

The roots exist only in the auxiliary frame. In the physical variables — the asymptotic charges an observer measures — the objects that survive are exactly the Galois invariants of the horizon cover.

For Kerr–Newman the cover is a quadratic and the distinction is soft: both $e_1 = S_+ + S_-$ and $e_2 = S_+S_-$ are rational in (M, J, Q) , and $\sqrt{\Delta}$ (one square root of an invariant) recovers the roots. We show in Section 2 that the entire classical left/right dictionary is the representation theory of the $\mathbb{Z}/2$ deck action on this quadratic: the even (invariant) sector is the left-moving sector, the odd sector is the right-moving sector, and the Galois-odd datum of the pair is the covariant $\Theta = ST$ — the orientation the invariant ring discards.

At four charges the distinction becomes a theorem. The seed mass obeys

$$4M = \sum_{i=1}^4 \sqrt{m^2 + N_i^2}, \quad N_i = 4Q_i, \quad (1)$$

a four-radical relation whose norm form is a *quintic* in $u = m^2$ over $\mathbb{Q}(M, Q_1, \dots, Q_4)$. We certify (Theorem 4.2) that its Galois group is S_5 : by the Abel–Ruffini theorem there is no closed-form radical expression for the seed mass in the physical variables. The parametric (cosh/sinh) presentation of four-charge thermodynamics is therefore not a convention but a necessity. We then show (Theorem 4.4) that the obstruction propagates to the entropy itself: for the static four-charge black hole the radical tower above the quintic collapses — $\prod_i (w_i + m) \cdot \prod_i (w_i - m) = \prod_i N_i^2$ is rational — so that $(S_+ + S_-)^2/\pi^2$ is a *primitive element* of the S_5 field, while S_+S_- is rational. The horizon entropies of a four-charge black hole cannot be written in radicals in the observables; their symmetric functions can, and the celebrated “universal” formulas of the literature are precisely the symmetric functions — *universality is Galois descent*.

The solvability threshold is sharp (Proposition 4.1): one and two charges are rational (the two-charge seed is, explicitly, a product of the four BPS walls), three charges give a solvable quartic, and four charges give the first S_5 . Finally, the Galois norm of the mass fiber factors completely over $\mathbb{Q}$ into the sixteen BPS-type linear forms (Proposition 5.1), identifying the marginal-stability walls of the charge lattice with the vanishing locus of a norm.

The quintic is only the base of the full horizon cover. Above each mass sheet lie square roots representing the axial sign, entropy sum, and entropy difference. Sections 6–9 determine this decorated layer. Signed contact divisors yield invertible valuation-parity matrices, proving that the conjugate square classes are independent. The resulting normal closures are maximal wreath products. In the rotating family, the entropy sum is independent of J , the entropy difference acquires a private prime, and the static relation between axial reflection and horizon swap becomes a rank-drop specialization at $J = 0$.

Relation to prior work. The thermodynamic dictionary of Section 2 is *known*: the sector first laws, Smarr formulas, negative inner temperatures, half-energy property, entropy product and

inversion appear in [3]; the hidden-conformal temperatures and central charge for Kerr–Newman in [8] (Kerr case [7]); the universal area products in [4, 5, 6]; the left/right formalism originates with [1, 2]. Our contribution there is the derivation principle: all of it follows from the $\mathbb{Z}/2$ Galois grading of one rational quadratic, with no geometric or conformal input. The number-theoretic study of *extremal* black holes — attractor arithmetic, class numbers, complex multiplication [14, 15] — concerns the abelian Galois theory of attractor values and is complementary to the questions here, which concern the non-abelian Galois theory of the *non-extremal cover*. To our knowledge, the algebraic-obstruction layer — the S_5 certification, the Abel–Ruffini statements for mass and entropy, the charge-count threshold, and the wall-factorization of the norm — has not appeared in the literature; a full prior-art search (INSPIRE full-text and expert consultation) is in progress, and these claims should be read with that caveat. Notably, [3] states the entropy–mass *inversion* formula for the quadratic case and develops the sector apparatus at length without raising the solvability question in the multi-charge case; a full-text search of that paper for the relevant algebraic vocabulary returns no occurrences.

Conventions. Units $G = \hbar = c = 1$; entropy $S = A/4$. For Kerr–Newman we use the fundamental relation in the form

$$M^2 = U(S, J, Q) = \frac{S}{4\pi} + \frac{\pi J^2}{S} + \frac{Q^2}{2} + \frac{\pi Q^4}{4S}, \quad (2)$$

with conjugate potentials $T = U_S/2M$, $\Omega = U_J/2M$, $\Phi = U_Q/2M$. All computer-verified statements are exact (symbolic or exact rational arithmetic); the scripts, their SHA-256 hashes, and the Dedekind witnesses are inventoried in Appendix A.

2 The Kerr–Newman horizon pair as a Galois cover

2.1 The quadratic, its invariants, and the horizon swap

Solving (2) for S at fixed (M, J, Q) gives the monic quadratic

$$S^2 - e_1 S + e_2 = 0, \quad e_1 = 2\pi(2M^2 - Q^2), \quad e_2 = \pi^2(4J^2 + Q^4), \quad (3)$$

whose roots are the outer and inner horizon entropies

$$S_{\pm} = \pi(2M^2 - Q^2 \pm 2\sqrt{\Delta}), \quad \Delta = M^4 - M^2 Q^2 - J^2, \quad (4)$$

with $e_1^2 - 4e_2 = 16\pi^2\Delta$. Both coefficients are *rational* in the asymptotic charges; e_2 is independent of the mass (the universal area product [4, 5] in entropy normalization). The Galois group of (3) over $\mathbb{Q}(M, J, Q)$ is $\mathbb{Z}/2$, realized geometrically as the *horizon swap* $S_+ \leftrightarrow S_-$, i.e. the deck transformation $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ of the double cover branched along the extremal locus $\Delta = 0$.

Three elementary facts organize the argument. Each is verified symbolically in Appendix A.

1. *Monodromy.* Transporting around $\Delta = 0$ swaps the sheets while e_1, e_2 are single-valued. The branch point is simple everywhere on the smooth extremal edge: $\Delta = 0$ and $\nabla\Delta = 0$ have no common solution away from $M = J = Q = 0$.
2. *Past the wall.* For $\Delta < 0$ (the over-extremal region) the pair becomes complex-conjugate, $S_- = \overline{S_+}$, with $\text{Re } S_{\pm} = e_1/2$ frozen at the extremal value and $\text{Im } S_{\pm} = \pm 2\pi\sqrt{|\Delta|}$. The individual root “goes complex”; the invariants cross the wall rationally. *The entropy never crosses; its symmetric functions do.*

3. *Vieta identities for the potentials.* On the two branches,

$$U_S(S_+) = +\frac{\sqrt{\Delta}}{S_+}, \quad U_S(S_-) = -\frac{\sqrt{\Delta}}{S_-}, \quad (5)$$

so that $T_{\pm} = \pm\sqrt{\Delta}/(2MS_{\pm})$: the inner-horizon temperature is negative in the Gibbsian convention, in agreement with [3].

2.2 The Galois-odd datum is $\Theta = ST$

Proposition 2.1 (Splitting–temperature identity). *Let $\Theta := ST$ on either branch, so that $\Theta_{\pm} = \pm\sqrt{\Delta}/2M$. Then*

$$S_+ - S_- = 4\pi\sqrt{\Delta} = 8\pi M \Theta_+.$$

Under the horizon swap Θ is odd, $\tau(\Theta_{\pm}) = -\Theta_{\pm}$, with invariant square $\Theta^2 = \Delta/4M^2$. The temperature itself is mixed — $\tau(T_+) = T_-$, an odd numerator over a branch-dependent denominator — and the two temperatures are Galois-conjugate over the observables, with rational symmetric functions (for $e_2 \neq 0$)

$$T_+ + T_- = -\frac{2\pi\Delta}{M e_2}, \quad T_+ T_- = -\frac{\Delta}{4M^2 e_2}.$$

At $J = Q = 0$ the identity returns the Schwarzschild value $T_+ = 1/8\pi M$ exactly.

The separation of the two sheets is the odd covariant. Since $(S_+ - S_-)^2 = e_1^2 - 4e_2$, the invariant ring sees Θ only through its square: the sign of Θ — outer versus inner, the arrow that distinguishes the horizons — is precisely the datum the Galois quotient forgets. Temperature inherits that arrow through $T = \Theta/S$, which is how the negative inner temperature of [3] arises, but T itself is not of pure parity: it is the odd covariant divided by a root. This is the organizing fact of the paper: *the thermodynamic quantities that are hard to define in the observables are the odd and mixed ones; the ones that “descend” are the invariants.*

2.3 The left/right dictionary from the grading

Define the even and odd sector entropies

$$S_L = \tfrac{1}{2}(S_+ + S_-) = \tfrac{1}{2}e_1 = \pi(2M^2 - Q^2), \quad S_R = \tfrac{1}{2}(S_+ - S_-) = 2\pi\sqrt{\Delta}, \quad (6)$$

and sector temperatures by thermodynamic conjugacy at fixed (J, Q) : $1/T_{L,R} := \partial_M S_{L,R}$. All of the following are exact identities (Appendix A):

$$T_L = \frac{1}{4\pi M} \quad (\text{mass only: } J \text{ and } Q \text{ drop out}), \quad T_R = \frac{\sqrt{\Delta}}{M e_1}, \quad (7)$$

$$\frac{1}{T_{\pm}} = \frac{1}{T_L} \pm \frac{1}{T_R} \quad (\text{harmonic composition}), \quad \left(\frac{T_R}{T_L}\right)^2 = 1 - \frac{4e_2}{e_1^2} = \frac{S_R^2}{S_L^2}. \quad (8)$$

The left/right asymmetry is the normalized discriminant of the quadratic. The sector split, the half-energy property of the left sector, and the harmonic composition agree with [3, 1]; the Cardy-consistent hidden-conformal temperatures of [8], $T_L^{\text{CFT}} = (2M^2 - Q^2)/4\pi J$ and $T_R^{\text{CFT}} = \sqrt{M^4 - J^2 - M^2 Q^2}/2\pi J$ with $c_L = c_R = 12J$, are $e_1/8\pi^2 J$ and $\sqrt{\Delta}/2\pi J$ in the present variables, and the Cardy formula returns exactly S_L and S_R (verified symbolically; Kerr case [7]). Extremal chirality — $S_R, T_R \rightarrow 0$ at $\Delta \rightarrow 0$, the purely left-moving limit of [10] — is, in this language, the death of the odd sector at the branch point.

Proposition 2.2 (Sector first laws and Smarr formulas; charge parity). *With sector potentials $\Omega_X = -T_X \partial_J S_X$, $\Phi_X = -T_X \partial_Q S_X$ one finds, exactly,*

$$\Omega_L \equiv 0, \quad \Phi_L = \frac{Q}{2M}, \quad M = 2T_L S_L + \Phi_L Q, \quad (9)$$

$$\Omega_R = \frac{J}{M(2M^2 - Q^2)}, \quad \Phi_R = \frac{MQ}{2M^2 - Q^2}, \quad M = 2T_R S_R + 2\Omega_R J + \Phi_R Q. \quad (10)$$

The left sector is a complete, non-rotating, charged thermodynamic system: the angular momentum couples exclusively to the odd sector. Charge behaves differently: Q appears in both e_1 and Δ , and couples to both sectors.

Sector first laws and Smarr formulas for the two-horizon system appear in [3]; we re-derive them here from the grading and record the following organizing observation, which to our knowledge has not been isolated as a principle (pending the picture-literature check, cf. [16]):

Observation 2.3 (Charge-parity principle). A conserved charge couples to the sectors according to which symmetric functions of the quadratic carry it. J appears only in the discriminant (equivalently only in e_2): it is *parity-pure*, purely right-moving, and $\Omega_L \equiv 0$. Q leaks into e_1 : it *straddles*. This is why the J -picture of the hidden-conformal correspondence is grading-natural (the odd sector’s circle is intrinsically the rotation angle, $c = 12J$ with no further choices), while charged pictures require auxiliary structure — the well-known arbitrary scale in their central charges is, in this language, the statement that no parity-pure circle exists for Q . On the Reissner–Nordström slice no charge is parity-pure at all.

2.4 Attractor as Vieta; duality as the symmetry of e_2

At the branch point the double root gives $S_{\text{ext}} = e_1/2 = \sqrt{e_2} = \pi\sqrt{4J^2 + Q^4}$: the extremal entropy is the square root of the constant term, mass-independent by inspection — the attractor-mechanism entropy formula [14] appearing as Vieta’s formula at the discriminant-zero locus. For dyonic Kerr–Newman ($Q^2 \rightarrow Q_e^2 + Q_m^2$) the electromagnetic duality group is exactly the symmetry group of e_2 : e_1, e_2, Δ are $SO(2)$ -invariant, the sector potentials transform as duality doublets, both sector Smarr formulas hold dyonically, and $T_L = 1/4\pi M$ is duality-blind (all verified exactly).

3 Four charges: the invariant persists

The four-charge generalization (the Cvetic–Youm class [11, 12]) is parametrized by a seed mass m , rotation a , and four boosts δ_i :

$$M = \frac{m}{4} \sum_i \cosh 2\delta_i, \quad Q_i = \frac{m}{4} \sinh 2\delta_i, \quad J = ma(\Pi_c - \Pi_s), \quad N_i := 4Q_i, \quad (11)$$

$$S_{\pm} = 2\pi m \left[m(\Pi_c + \Pi_s) \pm \sqrt{m^2 - a^2}(\Pi_c - \Pi_s) \right], \quad \Pi_c = \prod_i \cosh \delta_i, \quad \Pi_s = \prod_i \sinh \delta_i. \quad (12)$$

Two exact anchors pin this presentation to Section 2: the zero-boost limit reproduces the Kerr quadratic, and the equal-charge diagonal reproduces the Kerr–Newman quadratic (3) with $Q = \sum_i Q_i$ (both verified symbolically). The second symmetric function generalizes exactly:

Proposition 3.1 (The invariant at four charges). *As an unconditional algebraic identity in the parametrization,*

$$e_2 = S_+ S_- = \pi^2 (4J^2 + N_1 N_2 N_3 N_4). \quad (13)$$

The Kerr–Newman Q^4 is the equal-charge diagonal of the four-fold product. The mass enters only through (J, N_i) ; with quantized charges the invariant lies on a lattice — the classical shadow of level matching. The static sectors are $S_{L,R} = 2\pi m^2 (\Pi_c \pm \Pi_s)$: the Galois-even/odd split is the cosh/sinh product split of the string-theoretic left/right excitation counting [1, 13].

4 The unsolvability theorems

4.1 The mass fiber and its norm

Eliminating the boosts from the parametrization yields the four-radical relation (1) for the seed mass. Its norm form

$$\mathcal{N}(u) = \prod_{\varepsilon \in \{\pm 1\}^4} \left(4M - \sum_i \varepsilon_i \sqrt{u + N_i^2} \right), \quad u = m^2, \quad (14)$$

is a polynomial in u of degree exactly 5, with leading coefficient $-2^{24} M^6$ independent of the charges (Appendix B): the two aligned sign patterns contribute one u , the four ± 2 -pairs contribute four, and the six balanced $(2, 2)$ patterns contribute the M^6 .

Proposition 4.1 (The charge ladder). *For n charges the analogous norm has u -degree $1, 1, 4, 5$ at $n = 1, 2, 3, 4$. In particular: for $n \leq 2$ the seed is a rational function of the observables — explicitly, at two charges,*

$$m^2 = \frac{4(M - Q_1 - Q_2)(M + Q_1 + Q_2)(M - Q_1 + Q_2)(M + Q_1 - Q_2)}{M^2}, \quad (15)$$

the product of the four BPS walls; at $n = 3$ the seed obeys an irreducible quartic (solvable by radicals, though we are not aware of the formula having been written); and $n = 4$ is the first quintic. The solvability threshold sits exactly at four charges.

4.2 Abel–Ruffini for the mass fiber

Theorem 4.2 (S_5 obstruction, seed mass). *Over $\mathbb{Q}(M, Q_1, \dots, Q_4)$ the polynomial $\mathcal{N}(u)$ is irreducible with Galois group S_5 . Consequently the seed mass-square $u = m^2$ of the non-extremal four-charge black hole admits no closed-form expression in radicals in the physical variables, and the boost parametrization cannot be inverted in radicals.*

Proof (computer-certified; Appendix A). By the specialization principle, the Galois group of any separable rational specialization embeds in the generic group; hence a single specialization with non-solvable group forces the generic group to be non-solvable, and, being a transitive subgroup of S_5 witnessing S_5 cycle types, to equal S_5 . At three independent rational points — including the integer charge vector $M = 15$, $Q = (1, 2, 3, 5)$, chosen *directly* rather than through the parametrization, above the BPS bound $M > \sum_i Q_i$ — the specialized quintic is separable and irreducible over $\mathbb{Q}$, the physical (all-plus branch) root lies on it, and Dedekind reduction exhibits Frobenius elements of cycle type $(2, 3)$ at $p = 53, 23, 31$ respectively. A $(2, 3)$ element has order six and exists in no transitive quintic group except S_5 . All five roots are real at each point (the quintics are totally real): the physical seed has four real Galois-conjugate siblings over the same observables. $\square$

4.3 Abel–Ruffini for the entropy

The entropy involves the boosts as well as the seed; a priori the obstruction of Theorem 4.2 might be bypassed. It is not: the tower collapses.

Lemma 4.3 (Collapse of the radical tower; static case). *Let $w_i = \sqrt{m^2 + N_i^2}$ and $u = m^2$. Then, exactly: (i) the branch-compatible signed radicals lie in the quintic field — there exist $g_i \in \mathbb{Q}[u]/(\mathcal{N})$ with $g_i^2 \equiv u + N_i^2$ and $\sum_i g_i \equiv 4M$ (certified as polynomial identities modulo $\mathcal{N}$); (ii) with $A = \prod_i (w_i + m)$, $B = \prod_i (w_i - m)$ one has $AB = \prod_i N_i^2 \in \mathbb{Q}$, so that $y := (S_+ + S_-)/\pi = \sqrt{A} + \sqrt{B}$ satisfies*

$$y^2 = \gamma(u) := 2 \left[e_4(w) + u e_2(w) + u^2 + \prod_i N_i \right] \in \mathbb{Q}[u]/(\mathcal{N}). \quad (16)$$

Theorem 4.4 (Entropy non-radical; static four-charge). *γ is a primitive element of the quintic field: its resolvent $\text{Res}_u(\mathcal{N}, X - \gamma)$ is an irreducible quintic whose Galois group is again S_5 (Dedekind witnesses of type $(2, 3)$ at $p = 17, 19, 23$). Since $y^2 = \gamma$, the field $\mathbb{Q}(M, \vec{Q})(S_+ + S_-)$ contains the quintic field and its Galois closure surjects onto S_5 : it is non-solvable. Hence, in the physical variables:*

1. $S_+ + S_-$ admits no radical expression; its exact degree is 10 (the polynomial $\text{Res}_u(\mathcal{N}, Y^2 - \gamma)$ is irreducible);
2. $S_+ S_- = \pi^2 \prod_i N_i$ is rational, so S_+ and S_- individually admit no radical expression, and neither do S_L and S_R ;
3. $S_L^2 = (\pi^2/4) \gamma$: the left-sector level count is itself a primitive element of the unsolvable field, while the level-matched combination $S_L^2 - S_R^2 = e_2$ is rational.

Remark 4.5 (Scope). The non-radicality of the sector temperatures $T_{L,R}$ is plausible by inheritance but is not certified here. The five sheets of the quintic are realized, at the certified points, by the all-plus sign pattern and the four single-flip patterns: the Galois orbit of a four-charge black hole over its own observables consists of the hole and its four one-charge-conjugate shadows. Single charge conjugations act on the base (e_1, e_2) data — they move between quintic sheets — while the deck transformation of each entropy pair remains $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$.

Remark 4.6 (Methodological corollary). The left/right apparatus of [1, 2, 3] is constructed by explicit manipulation of $S_{\pm}$ in the boost frame. Theorems 4.2 and 4.4 show that at four charges this construction cannot be transported to the observables even in principle: the roots, the sector entropies, and the individual level counts are not radical functions of $(M, \vec{Q})$. The quantities of the apparatus that are expressible in the observables are precisely the Galois invariants — the area product, the level-matched difference, the wall data of Section 5. *Universality is descent: universal formulas are universal because they are norms.*

5 The norm and the walls

Proposition 5.1 (Wall factorization of the norm). *The product of the five branches of the mass fiber is, exactly,*

$$\prod_{k=1}^5 u_k = \frac{256}{M^6} \prod_{\varepsilon \in \{\pm 1\}^4} (M + \varepsilon_1 Q_1 + \varepsilon_2 Q_2 + \varepsilon_3 Q_3 + \varepsilon_4 Q_4), \quad (17)$$

with leading coefficient $\text{lc}(\mathcal{N}) = -2^{24}M^6$ independent of the charges. Equivalently $\mathcal{N}(0) = 4^{16} \prod_{16} (M \pm Q_1 \pm Q_2 \pm Q_3 \pm Q_4)$: the constant term of the quintic is the black hole evaluated at its own BPS limit ($m \rightarrow 0$), and the Galois norm vanishes exactly on the union of the sixteen marginal-stability walls. The norm is positive throughout the physical chamber $M > \sum_i |Q_i|$ (consistently with the totally real quintics of Theorem 4.2) and changes sign across walls: the chamber structure of the charge lattice is read off one coefficient.

Thus the invariant tower is, floor by floor, *more* rational than solvability requires: the roots need S_5 ; the quadratic invariant $e_2 = \pi^2(4J^2 + \prod N_i)$ is rational and quantized; and the quintic's top invariant does not merely remain rational — it splits completely over $\mathbb{Q}$ into the BPS walls. What the Galois quotient keeps is not just rational; it is rational *and physical*. At two charges the wall product *is* the seed (Proposition 4.1); at four charges it survives only as the norm of five entangled sheets.

6 Conjugate Kummer modules and wreath lifts

Let K/F be finite separable of degree d , let E/F be its normal closure, put $G = \text{Gal}(E/F)$, and let $I = \text{Hom}_F(K, E)$. For nonzero $r_1, \dots, r_s \in K$, write $r_{\alpha,i} = i(r_\alpha)$ and set

$$H = K(\sqrt{r_1}, \dots, \sqrt{r_s}), \quad L = E(\sqrt{r_{\alpha,i}} : \alpha \leq s, i \in I). \quad (18)$$

The conjugate Kummer module is the image W of

$$\vartheta : \mathbf{F}_2^s \otimes \mathbf{F}_2[I] \longrightarrow E^*/E^{*2}, \quad e_{\alpha,i} \longmapsto [r_{\alpha,i}]. \quad (19)$$

Theorem 6.1 (Conjugate Kummer module). *L is the normal closure of H/F , and restriction gives*

$$1 \longrightarrow W^\vee \longrightarrow \text{Gal}(L/F) \longrightarrow G \longrightarrow 1. \quad (20)$$

If all sd conjugate square classes are independent, then

$$\text{Gal}(L/F) \cong (C_2^s)^I \rtimes G = C_2^s \wr_I G, \quad |\text{Gal}(L/F)| = 2^{sd}|G|. \quad (21)$$

Proof. Every F -embedding of H sends $\sqrt{r_\alpha}$ to a signed square root of some $r_{\alpha,i}$, so its normal closure contains L . Conversely the displayed generators are permuted up to sign by every F -embedding, hence L/F is normal. Kummer theory over E identifies the kernel with $W^\vee$. In the independent case the signed-permutation representation into $(C_2^s)^I \rtimes G$ is injective and the orders agree. $\square$

A discrete valuation v of E supplies the functional $[x] \mapsto v(x) \bmod 2$. Thus a parity matrix

$$A = (v_q(r_{\alpha,i}) \bmod 2) \quad (22)$$

of full column rank proves independence. If s divisor types are supported on one distinguished sheet and their parity rows form $B \in \text{GL}_s(\mathbf{F}_2)$, transitivity of G transports them to every sheet and produces $B \otimes I_d$.

7 The degree-20 crown and static normal closure

Return to $F = \mathbb{Q}(M, N_1, N_2, N_3, N_4)$, $K = F(u)$, and $P = \prod_i N_i$. Put

$$\alpha = e_4(w) + ue_2(w) + u^2, \quad \beta = e_3(w) + ue_1(w), \quad \gamma = 2(\alpha + P). \quad (23)$$

Since $\prod_i (w_i \pm m) = \alpha \pm m\beta$, one obtains

$$\alpha^2 - u\beta^2 = P^2, \quad \gamma(\gamma - 4P) = 4u\beta^2. \quad (24)$$

For a sign vector ε , let $\ell_\varepsilon = 4M - \sum_i \varepsilon_i N_i$ and $s_\varepsilon = \prod_i \varepsilon_i$. At its signed contact component, $u = 0$ and $w_i = \varepsilon_i N_i$. Away from $\sum_i \varepsilon_i / N_i = 0$, the valuation parities are

s_ε	$v(u)$	$v(\gamma)$	$v(\gamma - 4P)$
+1	1	0	1
-1	1	1	0

Theorem 7.1 (Degree-20 crown). *Generically $[u]$ and $[\gamma]$ are independent in K^*/K^{*2} . Hence*

$$[K(\sqrt{u}, \sqrt{\gamma}) : K] = 4, \quad [K(\sqrt{u}, \sqrt{\gamma}) : F] = 20. \quad (25)$$

With $\widehat{S}_\pm = S_\pm/\pi$, either horizon generates this degree-20 ordered-horizon field.

Proof. An even contact valuation proves that u and $u\gamma$ are nonsquares; an odd contact valuation proves that γ is a nonsquare. If $y = \widehat{S}_+ + \widehat{S}_-$ and $z = \widehat{S}_+ - \widehat{S}_-$, then $y^2 = \gamma$ and $z = 2\beta\sqrt{u}/y$. The primitivity $K = F(\gamma)$ from Theorem 4.4, together with $\widehat{S}_+\widehat{S}_- = P \in F$, shows that either horizon recovers the other and then the entire field. $\square$

Let u_i, γ_i be the five conjugates in the S_5 splitting field E . Exact even and odd wall witnesses show that the selected contacts are not components of the quintic discriminant, hence their valuations are unramified in E/F . Their S_5 -orbits give

$$V_{\text{st}} = \begin{pmatrix} I_5 & 0 \\ I_5 & I_5 \end{pmatrix} \quad (26)$$

on the columns $(u_1, \dots, u_5 \mid \gamma_1, \dots, \gamma_5)$.

Theorem 7.2 (Static wreath closure). *The ten conjugate square classes are independent and*

$$\text{Gal}(\widetilde{H}/F) \cong C_2^2 \text{ wr } S_5, \quad |\text{Gal}(\widetilde{H}/F)| = 2^{10} \cdot 120 = 122880. \quad (27)$$

The action has five mass-sheet blocks of size four. In each block, V_4 acts regularly on

$$\{\widehat{S}_+, \widehat{S}_-, -\widehat{S}_+, -\widehat{S}_-\}.$$

8 Rotation and the third Kummer channel

Let J be transcendental over F , put $F_J = F(J)$, $K_J = K(J)$, and define

$$\delta_J = \gamma - 4P - 16J^2. \quad (28)$$

The physical parameter map is generically dominant: at $a = 0$ its static Jacobian has rank five and $\partial J/\partial a = m(\Pi_c - \Pi_s) \neq 0$ generically. Thus the purely transcendental model is the generic rotating observable field.

Theorem 8.1 (Rank jump under rotation). *The classes $[u], [\gamma], [\delta_J]$ are independent in K_J^*/K_J^{*2} . Consequently*

$$H_J = K_J(\sqrt{\gamma}, \sqrt{\delta_J}), \quad [H_J : F_J] = 20, \quad (29)$$

$$L_J = K_J(\sqrt{u}, \sqrt{\gamma}, \sqrt{\delta_J}), \quad [L_J : F_J] = 40. \quad (30)$$

At $J = 0$, $[\delta_0] = [u][\gamma]$ and

$$K(\sqrt{\gamma}, \sqrt{\delta_0}) = K(\sqrt{u}, \sqrt{\gamma}). \quad (31)$$

The degree-eight augmented Kummer algebra specializes to two isomorphic degree-four components; choosing a compatible sign selects the static crown.

Proof. Pure transcendental extension preserves the quintic and the old square classes. By (24), $[\gamma - 4P] = [u][\gamma]$, hence $\gamma - 4P$ is a nonsquare. Therefore $\delta_J = (\gamma - 4P) - 16J^2$ is irreducible in $K[J]$ and its height-one prime has odd valuation on δ_J but zero valuation on the two constant classes. Kummer theory gives the degrees. At $J = 0$, $\sqrt{\delta_0} = 2\beta\sqrt{u}/\sqrt{\gamma}$, proving (31); over the static crown the final quadratic splits. $\square$

In the boost frame,

$$\frac{S_+ + S_-}{\pi} = 4m^2(\Pi_c + \Pi_s), \quad \frac{S_+ S_-}{\pi^2} = P + 4J^2, \quad (32)$$

so $y^2 = \gamma$ remains independent of rotation and $z^2 = \delta_J$. Since the product is in F_J , either rotating horizon again generates H_J .

Over $E(J)$ put $\delta_i = \gamma_i - 4P - 16J^2$. The static contact valuations extend as Gauss valuations and have zero δ -parity. Each δ_i is squarefree in J , and distinct sheets have coprime δ_i because $\delta_i - \delta_j = \gamma_i - \gamma_j \neq 0$. Choosing one private prime per sheet gives

$$V_{\text{aug}} = \begin{pmatrix} I_5 & 0 & 0 \\ I_5 & I_5 & 0 \\ 0 & 0 & I_5 \end{pmatrix}, \quad V_{\text{hor}} = \begin{pmatrix} I_5 & 0 \\ 0 & I_5 \end{pmatrix}. \quad (33)$$

Theorem 8.2 (Rotating wreath closures). *The ordered-horizon and augmented rotating normal closures satisfy*

$$\text{Gal}(\tilde{H}_J/F_J) \cong C_2^2 \text{ wr } S_5, \quad |\text{Gal}(\tilde{H}_J/F_J)| = 122880, \quad (34)$$

$$\text{Gal}(\tilde{L}_J/F_J) \cong C_2^3 \text{ wr } S_5, \quad |\text{Gal}(\tilde{L}_J/F_J)| = 3932160. \quad (35)$$

Proof. Both matrices in (33) are invertible. Theorem 6.1 applies with the base group S_5 . $\square$

9 Inertia and the realized branch stratification

The valuation proof also classifies tame ramification. Locally write $r_{\alpha,i} = \varpi^{m_{\alpha,i}} \epsilon_{\alpha,i}$ with a unit $\epsilon_{\alpha,i}$. Away from residue characteristic two, generic divisorial Kummer inertia is cyclic of order two and flips precisely the square roots with $m_{\alpha,i}$ odd. At a normal-crossing intersection, its rank is the $\mathbf{F}_2$ -rank of the incident parity rows.

For $(a; \sigma) \in (C_2^s)^5 \rtimes S_5$, each cycle c of σ carries the color $\kappa_c = \sum_{i \in c} a_i \in C_2^s$. Cycle lengths and cycle sums determine the wreath conjugacy class. A cycle of length ℓ has order ℓ when $\kappa_c = 0$ and 2ℓ otherwise. The generic divisorial catalogue is therefore

Locus	Parity/color	Generic inertia
simple mass branch	uncolored transposition	C_2
even signed contact	axial $(1, 0, 0)$	C_2
odd signed contact	axial-sum $(1, 1, 0)$	C_2
rotating difference $\delta_i = 0$	difference $(0, 0, 1)$	C_2
reciprocal sub-balance	colored transposition	C_4
$J = 0$	Kummer relation divisor	unramified rank drop
$J = \infty$, generic weighted infinity	even parity	unramified
$N_i^2 = N_j^2$	normalized sheet crossing	unramified

At a collision stratum with local permutation group $P_0 \leq S_5$, let U be the P_0 -stable span of the incident parity rows. The local group is the subgroup generated inside $U \rtimes P_0$. This separates the intrinsic inertia theorem from the geometric question of which colored partitions are realized.

That realization question was computed on the incidence cover

$$X : \quad w_i^2 - u - N_i^2 = 0 \quad (1 \leq i \leq 4), \quad \sum_i w_i = 4M. \quad (36)$$

Its relative Jacobian determinant is $8e_3(w)$. Macaulay2 gives

Stratum	codimension in X	degree
mass ramification I_R	1	48
even/odd contact	1	2
rotating difference I_Z	1	64
$I_R + I_Z$	2	192
even/odd reciprocal sub-balance	2	6
contact $+I_Z$	2	8
$I_R + \text{contact} + I_Z$	3	24

Theorem 9.1 (Realization-poset certificate). *After saturation by $2MP \prod_{a < b} (N_a^2 - N_b^2)$, every named stratum is nonempty with the displayed codimension and degree. The difference ideal*

$$I_Z = I_X + (P + 8J^2 - u^2 - ue_2(w) - e_4(w)) \quad (37)$$

is prime, reduced, and irreducible over $\mathbb{Q}$; its minimal-prime and primary decompositions each consist of I_Z alone. The intersection $I_R + I_Z$ has a single irreducible support. Even contact-difference intersections are supported on $J = 0$ with multiplicity two.

The computation also shows why the discriminant of the projected quintic is not the branch divisor: at $N_1 = N_2$ it acquires five extra squared factors from unramified crossings, whereas the true Jacobian branch image contains only the genuine factors. Ramification must be computed on the normalized incidence cover, not inferred from the projected eliminant discriminant.

10 Discussion and open problems

The picture that emerges is a division of black-hole thermodynamics into an invariant layer and an odd/root layer, separated — at four charges — by a non-solvable wall. The invariant layer (area products, level matching, wall data, the left sector's half-energy) descends to the observables and is

where the literature’s “universal” results live; the root layer (individual entropies, individual level counts, the sign of the temperature) exists only in the duality frame, and the solution-generating map is, in a precise sense, radical-one-way. The classical dictionary of the hidden-conformal program is reproduced by the grading alone, with no conformal input; the classical checks therefore carry no discriminating power about the existence of a dual CFT, and the burden of evidence rests on the genuinely quantum observables.

The closure problem posed by the original draft is now resolved: the decorated cover is a degree-20 crown with full C_2^2 wr S_5 normal closure, and rotation supplies a controlled third Kummer channel. Open problems are now higher in the tower: *(i)* extend the contact-parity realization and maximal wreath theorem to every charge number for which the symmetric base monodromy is known; *(ii)* determine the lower symmetric functions of the mass fiber; *(iii)* certify the sector-temperature fields; *(iv)* interpret the Kummer characters and inertia colors microscopically; *(v)* classify compactified higher-collision strata; and *(vi)* treat five-dimensional and cosmological-constant families.

A Computational certificates

All results marked as verified are exact: symbolic identities in **sympy**, or exact rational arithmetic with numerics used only as scaffolding (every numerically-guided step is discharged by an exact polynomial identity, e.g. modulo $\mathcal{N}$). Scripts (SHA-256 prefixes):

Script	Contents	SHA-256 (prefix)
<code>four_charge_crown.py</code>	Prop. 3.1; anchors; extremal chirality	8ff83aef7bc0
<code>abel_ruffini_black_holes.py</code>	Thm. 4.2; ladder points; witnesses	675b3ced84f0
<code>entropy_sum_resolvent.py</code>	Lem. 4.3; Thm. 4.4; degree 10	7f778cb81dce
<code>horizon_wreath_inertia_model.m2</code>	Thm. 9.1; incidence poset	d7940c05f902

Kerr–Newman identities of Section 2 (Vieta identities, splitting–temperature, sector laws, harmonic and ratio laws, Cardy match, attractor, duality) were verified in-session as standalone symbolic runs; the wall factorization of Proposition 5.1 was verified exactly at four rational points together with the charge-independence of the leading coefficient.

Dedekind witnesses (Theorem 4.2). Point A: $M = 3$, $Q = (\frac{1}{3}, \frac{1}{2}, \frac{2}{5}, \frac{3}{7})$; irreducible; witness $p = 53$, type $(2, 3)$; 5-cycles at $p = 41, 43$. Point B: $M = 15$, $Q = (1, 2, 3, 5)$; irreducible; witness $p = 23$; 5-cycles at $p = 29, 43, 47$. Point C: $M = 2$, $Q = (\frac{2}{7}, \frac{5}{9}, \frac{1}{4}, \frac{4}{11})$; irreducible; witness $p = 31$. All quintics separable and totally real; the physical root verified on each to residuals $\leq 10^{-15}$ (exact irreducibility making the identification rigorous).

Witnesses (Theorem 4.4, point B). Exact certificates modulo $\mathcal{N}$: $g_i^2 \equiv u + N_i^2$, $\sum g_i \equiv 4M$, $\alpha^2 - u\beta^2 \equiv (\prod N_i)^2$. Resolvent irreducible; witnesses $p = 17, 19, 23$, all type $(2, 3)$; $E(Y)$ of degree 10 irreducible over $\mathbb{Q}$; physical cross-check exact to 80 digits. Branch patterns of the five sheets: $(+ + + +)$ and the four single-flips.

Macaulay2 realization certificate. The master script was run under Macaulay2 1.25.11. All model assertions pass: $\text{codim } I_X = 5$, $\dim(S/I_X) = 6$, and the relative Jacobian determinant reduces to $8e_3(w)$. The full symbolic commands `minimalPrimes` `IZ` and `primaryDecomposition`

IZ each return one component, equal to I_Z , of ambient codimension six and degree 64. The raw stage outputs, scripts, and run report are included with the source package. The projected collision calculation is retained as an explicit warning against replacing the normalized branch ideal by a quintic discriminant.

B Degree count and the two-charge closed form

For large u each factor of (14) behaves as $4M - (\sum_i \varepsilon_i)\sqrt{u} + O(u^{-1/2})$. Pairing ε with $-\varepsilon$: the aligned pair contributes $16M^2 - 16u + \dots$, each of the four $|\sum \varepsilon| = 2$ pairs contributes $16M^2 - 4u + \dots$, and each of the three balanced pairs contributes $16M^2 + O(1)$; half-integer powers cancel because $\mathcal{N}$ is a polynomial. Hence $\deg_u \mathcal{N} = 1 + 4 + 0 = 5$ and $\text{lc} = (-16)(-4)^4(4M)^6 = -2^{24}M^6$. The same count gives degrees 1, 1, 4 at $n = 1, 2, 3$ charges; at $n = 2$ the norm is linear in u and solving it yields the wall-product closed form of Proposition 4.1.

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